Calculus A(2) Spring 2021 Midterm Exam

Name and Student ID: ____________________

1. (40 points) For each of the following questions choose **one answer** from A to D.

(a) A sequence $\{a_n\}_{n=1}^{\infty}$ converges to L. Which of the following statements is true?

A. There is always n such that $a_n = L$. ← For example, we can have $a_n < L$, $\forall n$
B. There is always n such that $a_n < L$ or $a_n > L$. ← We can have $a_n = L$, $\forall n$
C. There is always n such that $L - 1 < a_n < L + 1$. True because a_n approaches L for large n.
D. None of the above

(b) Which of the following statements is true about the infinite series $\sum_{n=1}^{\infty}(-1)^n/(n^2+n)$?

A. The series converges absolutely.
B. The series converges conditionally.
C. The series diverges.
D. The convergence of the series depends on how the summation is performed.

$|a_n| = \frac{1}{n^2+n} < \frac{1}{n^2}$.

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges $\Rightarrow \sum_{n=1}^{\infty} |a_n|$ converges.

(c) What is the coefficient of the second (nonvanishing) term in the Taylor series generated by $\sin x$ at $x = \pi/4$?

A. 1
B. $1/\sqrt{2}$
C. $-1/3!$
D. None of the above

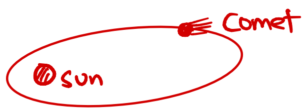
Taylor series:

$f(a) + f'(a)(x-a) + \cdots$

$f(x) = \sin x$, $a = \frac{\pi}{4}$

$\Rightarrow f(a) = \frac{1}{\sqrt{2}} \neq 0$, $f'(a) = \frac{1}{\sqrt{2}} \neq 0$.

(d) Halley's Comet has an orbital period of 75.32 years. Which of the following best describes its orbit?



A. A circle
B. An ellipse
C. A parabola
D. A hyperbola

$\left.\begin{matrix}\text{C}\\\text{D}\end{matrix}\right\}$ can't be true because the comet comes back every 75.32 years

(e) A point in a plane has polar coordinates $(r, \theta) = (1, 125\pi/2)$. What is the Cartesian coordinate (x, y) of this point?

A. $(1, 0)$
B. $(0, 1)$
C. $(-1, 0)$
D. $(0, -1)$

$\frac{125\pi}{2} = \underbrace{2\pi \times 31}_{\text{doesn't matter}} + \frac{\pi}{2}$.

(f) What is the dot product $\mathbf{u} \cdot \mathbf{v}$ of $\mathbf{u} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = -(\sin t)\mathbf{i} + (\cos t)\mathbf{j}$?

A. 0
B. 1
C. $-(2\cos t)\mathbf{i} - (2\sin t)\mathbf{j} + \mathbf{k}$
D. None of the above

$\mathbf{u} \cdot \mathbf{v} = (\cos t)(-\sin t) + (\sin t)(\cos t)$
$= 0$

(g) What is the cross product $\mathbf{u} \times \mathbf{v}$ of $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + 3\cancel{\mathbf{j}}$ $\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} + 6\mathbf{j} + 9\cancel{\mathbf{j}}$?

But this doesn't change the answer...

$\mathbf{u} \times \mathbf{v} = \mathbf{0}$ since $\mathbf{u}$ & $\mathbf{v}$ are parallel

A. 0 (scalar)
B. $\mathbf{0}$ (vector)
C. 42
D. None of the above

(h) Which of the following equations describes the line in space that is parallel to the vector $\mathbf{i} + \mathbf{j}$ and goes through point $(x, y, z) = (1, 1, 1)$?

A. $\mathbf{r}(t) = (1 + 2t)\mathbf{i} + (1 + 2t)\mathbf{j} + \mathbf{k}, \quad -\infty < t < \infty$ $\rightarrow \begin{cases} \mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k} \checkmark \\ \mathbf{r}'(t) = 2\mathbf{i} + 2\mathbf{j} \propto \mathbf{i} + \mathbf{j} \checkmark \end{cases}$
B. $\mathbf{r}(t) = (1 + t)\mathbf{i} + (1 + t)\mathbf{j} + (1 + 2t)\mathbf{k}, \quad -\infty < t < \infty$
C. $\mathbf{r}(t) = (1 + t)\mathbf{i} + (1 + t)\mathbf{j} + t\mathbf{k}, \quad -\infty < t < \infty$
D. None of the above

wrong $\mathbf{r}'$

wrong $\mathbf{r}'$, wrong $\mathbf{r}(0)$

(i) A point P_0 lies in a plane in space, and two different vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are both perpendicular to this plane. Which of the following equations describes this plane (as the set of points P satisfying it)?

A. $\mathbf{n}_1 \cdot \overrightarrow{P_0P} = \mathbf{n}_2 \cdot \overrightarrow{P_0P}$.
B. $\mathbf{n}_1 \times \overrightarrow{P_0P} = \mathbf{n}_2 \times \overrightarrow{P_0P}$.
C. $(\mathbf{n}_1 \times \mathbf{n}_2) \cdot \overrightarrow{P_0P} = 0$.
D. None of the above

$\mathbf{n}_1 \neq \mathbf{n}_2 \Rightarrow \mathbf{n}_1 - \mathbf{n}_2 \neq \mathbf{0}$ and perpendicular to the plane

$\Rightarrow (\mathbf{n}_1 - \mathbf{n}_2) \cdot \overrightarrow{P_0P} = 0$ describes the plane.

$\overrightarrow{P_0P}$ parallel to $\mathbf{n}_1$ & $\mathbf{n}_2$

$\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$, so satisfied by all points.

(j) A particle is moving with constant acceleration along the circle $x^2 + y^2 = 1$ in space. Which of the following statements is true about the position vector $\mathbf{r}$ and the velocity $\mathbf{v}$ of the particle?

A. $\mathbf{r} \cdot \mathbf{v} = 0$.
B. $\mathbf{r} \times \mathbf{v} = 0$. $\leftarrow \mathbf{r} \times \mathbf{v} \neq \mathbf{0}$
C. $|\mathbf{v}|$ is constant.
D. None of the above is true.

$|\mathbf{r}|^2 = \mathbf{r} \cdot \mathbf{r} = 1$

$\Rightarrow \frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = 2\mathbf{r} \cdot \mathbf{v} = 0$

(6+6 +6 pts) 2. (18 points) Find the interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{\sqrt{2n}-\sqrt{n-1}}{n} x^n .$$

① Use Ratio Test :

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n}{n+1} \cdot \frac{\sqrt{2n+2}-\sqrt{n}}{\sqrt{2n}-\sqrt{n-1}} \cdot \frac{x^{n+1}}{x^n}\right|$$

$$= \left|\frac{1}{1+\frac{1}{n}} \cdot \frac{\sqrt{2+2/n}-\sqrt{1}}{\sqrt{2}-\sqrt{1-1/n}} \cdot \frac{x^{n+1}}{x^n}\right|$$

$$\xrightarrow{n\to\infty} \left|\frac{1}{1} \cdot \frac{\sqrt{2}-1}{\sqrt{2}-1} \cdot \frac{x^{n+1}}{x^n}\right|$$

$$= |x| .$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ converges absolutely

if $|x|<1$, or $-1<x<1$

② Case $x=1$.

$$\sqrt{2n}-\sqrt{n-1} > \sqrt{2n}-\sqrt{n} = \overbrace{(\sqrt{2}-1)}^{>0}\sqrt{n}$$

$$\Rightarrow a_n > \frac{(\sqrt{2}-1)\sqrt{n}}{n} = \frac{\sqrt{2}-1}{n^{1/2}}$$

p-series with $p<\frac{1}{2}$ diverges

$\Rightarrow \Sigma a_n$ diverges

③ Case $x=-1$

$$\Sigma a_n = \Sigma (-1)^n \underbrace{\frac{\sqrt{2n}-\sqrt{n-1}}{n}}_{>0}$$

$\frac{\sqrt{2n}-\sqrt{n-1}}{n} \to 0$, it converges by Alternating Series Test.

Interval of convergence : $[-1, 1)$

(4+4 +4 pts) 3. (12 points) Calculate

$$\lim_{x\to 0} \frac{1}{7!} \cdot \frac{[\ln(1+x)]^7}{\sin x - x + \frac{x^3}{6} - \frac{x^5}{120}} .$$

① By Taylor series of $\ln(1+x)$,

$$[\ln(1+x)]^7 = [x+\cdots]^7 = x^7+\cdots$$

higher powers of x

② By Taylor series of $\sin x$,

$$\sin x - x + \frac{x^3}{6} - \frac{x^5}{120} = -\frac{x^7}{7!} + \cdots$$

③

$$\lim_{x\to 0} \frac{1}{7!} \frac{[\ln(1+x)]^7}{\sin x - x + \frac{x^3}{6} - \frac{x^5}{120}} = \lim_{x\to 0} \frac{1}{7!} \frac{x^7+\cdots}{-\frac{x^7}{7!}+\cdots} = -1 .$$

4. (12 points) Consider the infinite series $\sum_{n=0}^{\infty} 3(-2\cos\theta)^n$.

(3+3 pts) (a) For what values of θ does the series converge?

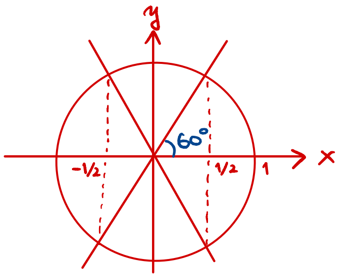
① Geometric series $\sum_{n=0}^{\infty} ar^n$ with $a=3$, $r=-2\cos\theta$.

$\Rightarrow$ Converges iff $|r|<1$

② $|r|<1 \iff -\frac{1}{2} < \cos\theta < \frac{1}{2}$

$\iff \frac{\pi}{3} < \theta < \frac{2\pi}{3}$ & $-\frac{2\pi}{3} < \theta < -\frac{\pi}{3}$

(or $\frac{4\pi}{3} < \theta < \frac{5\pi}{3}$)

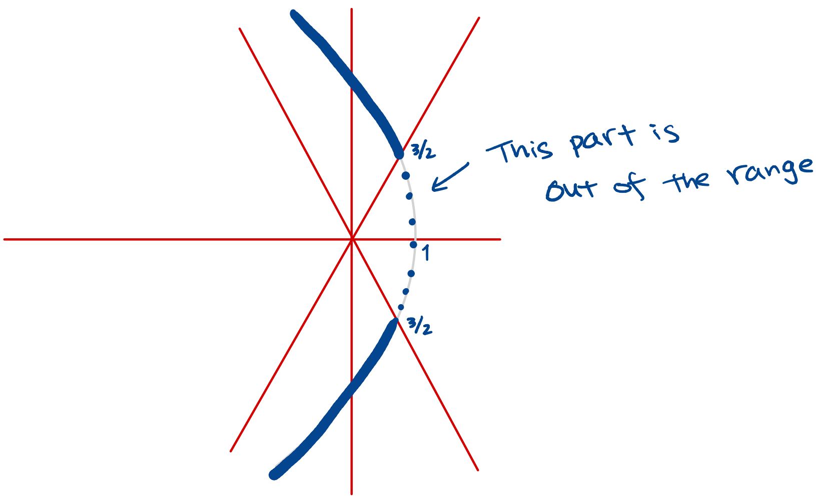


(3+3pts) (b) For the values of θ found in (a), let $f(\theta)$ be the sum of the series. Sketch the graph of the equation $r = f(\theta)$ in polar coordinates (r, θ).

① If converges, the sum is

$$f(\theta) = \frac{a}{1-r} = \frac{3}{1+2\cos\theta}$$

② $r = f(\theta)$ is a hyperbola:



5. (18 points) Alice is in a spaceship that is moving with velocity $\mathbf{v}(t) = -2\sin(2t)\mathbf{i} + \mathbf{j} + 4\cos(2t)\mathbf{k}$ at time t.

(3+3 pts) (a) Alice is at point $(x, y, z) = (1, 1, 1)$ at $t = 0$. Find the acceleration $\mathbf{a}(t)$ and the position $\mathbf{r}(t)$ of Alice at time t.

① $\mathbf{a}(t) = \frac{d}{dt}\mathbf{v}(t) = -4\cos(2t)\,\mathbf{i} + 8\sin(2t)\,\mathbf{k}$

② $\mathbf{r}(t) = \int \mathbf{v}(t)\,dt = \cos(2t)\,\mathbf{i} + t\,\mathbf{j} + 2\sin(2t)\,\mathbf{k} + \mathbf{C}$

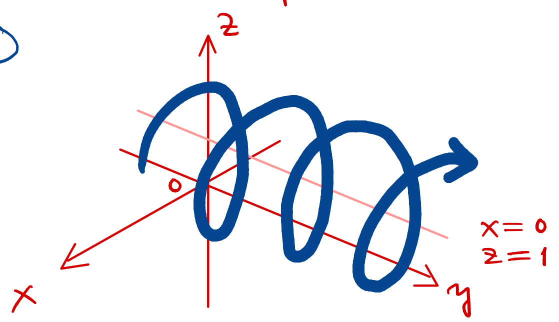
$\mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k} \Rightarrow \mathbf{C} = \mathbf{j} + \mathbf{k}$

$\Rightarrow \mathbf{r}(t) = \cos(2t)\,\mathbf{i} + (t+1)\,\mathbf{j} + (2\sin(2t)+1)\,\mathbf{k}$

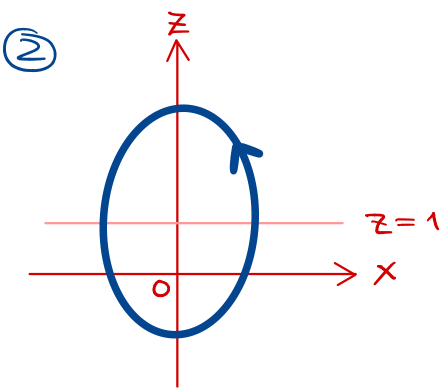
(3+3 pts) (b) Sketch the trajectory of Alice, and sketch its projection to the xz-plane.

It's like a helix, but the projection to xz-plane is an ellipse.

①



②



(3+3 pts) (c) Bob observes Alice from another spaceship moving with velocity $\mathbf{w}(t) = \sin(2t)\mathbf{i} + \cos(2t)\mathbf{k}$. Find the arc length that Alice travels from $t = 0$ to $t = 1$, as measured by Bob (that is, measured in a coordinate system whose origin is Bob's location).

① Alice's velocity as measured by Bob:

$\mathbf{v}(t) - \mathbf{w}(t) = -3\sin(2t)\,\mathbf{i} + \mathbf{j} + 3\cos(2t)\,\mathbf{k}$

② We have $|\mathbf{v} - \mathbf{w}| = \sqrt{3^2 + 1} = \sqrt{10}$

$\Rightarrow \int_0^1 |\mathbf{v} - \mathbf{w}|\,dt = \sqrt{10}$